

HW 09

lunes, 20 de julio de 2026

23:26

Problem 9.18

a) $V_1 = 60 \angle 15^\circ \text{ V}$, $\omega = 1$

$$V_1(t) = 60 \cos(t + 15^\circ) \text{ V}$$

b) $V_2 = 6 + j8 \text{ V}$, $\omega = 40$

$$|V_2| = \sqrt{6^2 + 8^2} = 10 \Rightarrow \angle = \tan^{-1}\left(\frac{8}{6}\right) = 53.13^\circ$$

$$V_2(t) = 10 \cos(40t + 53.13^\circ) \text{ V}$$

c) $I_1 = 2.8 e^{-j\pi/3} \text{ A}$, $\omega = 377$

$$-\pi/3 \text{ rad} \equiv -60^\circ$$

$$\Rightarrow i_1(t) = 2.8 \cos(377 - 60^\circ) \text{ A}$$

d) $I_2 = -0.5 - j1.2 \text{ A}$, $\omega = 10^3$

$$|I_2| = \sqrt{0.5^2 + 1.2^2} = 1.3$$

$$\angle = -180^\circ + \arctan\left(\frac{1.2}{0.5}\right) = -112.62^\circ$$

$$\Rightarrow i_2(t) = 1.3 \cos(10^3 t - 112.62^\circ)$$

Problem 9.25

a) $2 \frac{di}{dt} + 3i = 4 \cos(2t - 45^\circ)$, $\omega = 2$

$$RHS = 4 \angle -45^\circ$$

$$(3 + j2\omega) I = 4 \angle -45^\circ$$

$$\Rightarrow (3 + j4) I = 4 \angle -45^\circ$$

$$I = \frac{4 \angle -45^\circ}{5 \angle 53.13^\circ} = 0.8 \angle -98.13^\circ A$$

$$i(t) = 0.8 \cos(2t - 98.13^\circ) A$$

$$b) 10 \int i dt + \frac{di}{dt} + 6i = 5 \cos(5t + 22^\circ), \omega = 5$$

$$\frac{10}{j\omega} I + j\omega I + 6I = 5 \angle 22^\circ$$

$$\frac{10}{j5} = -j2, j\omega = j5$$

$$(6 + j3) I = 5 \angle 22^\circ \Rightarrow I = \frac{5 \angle 22^\circ}{6.708 \angle 26.57^\circ}$$

$$I = 0.745 \angle -4.57^\circ A$$

$$i(t) = 0.745 \cos(5t - 4.57^\circ) A$$

Problem 9.30

Resistor ($R = 40 \text{ k}\Omega$)

$$I_R = \frac{V}{R} = \frac{100 \angle 20^\circ}{40000} = 2.5 \angle 20^\circ \text{ mA}$$

$$i_R(t) = 2.5 \cos(60t + 20^\circ) \text{ mA}$$

$$\text{Capacitor } (C = 50 \mu\text{F}): I_C = \underbrace{j\omega C}_{60} V$$